THE NEGATIVE-VISCOSITY EFFECT IN FERROFLUIDS

A Complete First-Principles Reconstruction of Every Equation

VOLUME V

The Averaged Torque, the Two-Scale Expansion,
and the Central Result:
Equations (SM-25)–(SM-34)

Where Volume IV ended. Volume IV decomposed the linearly polarised field into two counter-rotating circular components, solved for each exactly, showed that a vortex Doppler-shifts their apparent frequencies to $\omega \mp \Omega$, and obtained from the difference of the two responses the transverse magnetisation

$$M_y = \Omega\tau_B\,\chi H_0 \cos^2\alpha\,\cos(\omega t - 2\alpha), \qquad \tan\alpha = \omega\tau_B. \tag{SM-23, SM-24}$$

It closed by previewing the reason the effect can change sign: the lag of M_y is 2α, and $\cos 2\alpha$ reverses at $\omega\tau_B = 1$.

Volume V executes that calculation. It computes the instantaneous torque (SM-25), separates the fast and slow time scales to obtain the averaged momentum equation (SM-26), extracts the rotational viscosity (SM-27), and arrives at the central result (SM-28) — the equation for which both papers were written. It then gives the high-frequency asymptote (SM-29), re-derives the same result by the authors' shorter second route (SM-30)–(SM-32), and closes the logical circle by computing the mean particle spin (SM-33) and recovering the master formula (SM-34) of Volume II. Chapter numbers resume at 39, page numbers at 142.

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Part X

The Alternating-Field Result

Chapter 39

Equation (SM-25): The Instantaneous Torque

$$\begin{aligned}\boldsymbol{M}\times\boldsymbol{H} &= -\Omega\tau_B\chi H_0^2\cos^2\alpha\,\cos\omega t\,\cos(\omega t-2\alpha)\\ &= -2\Omega\eta\varphi\xi^2\cos^2\alpha\big(\cos^2\omega t\,\cos 2\alpha+\sin\omega t\,\cos\omega t\,\sin 2\alpha\big).\end{aligned}\tag{SM-25}$$

39.1 Context

Volume IV produced a magnetisation with a component perpendicular to the field. A magnetisation not parallel to the field exerts a torque, by Eq. (4.4.2), and that torque is the mechanical agent of the whole effect: through (SM-12) it enters the momentum equation, and through (SM-11) it determines the particle spin. Equation (SM-25) computes it.

The result is written twice on purpose. The first line is the direct product; the second re-expresses it in the units natural to a viscosity ($\eta\varphi$) and separates the two time dependences, so that the averaging of the next chapter can be performed by inspection.

39.2 Symbols

Everything has appeared. Recall in particular

$$\xi=\frac{mH_0}{kT\sqrt{2}},\qquad \chi=\frac{nm^2}{3kT},\qquad \tau_B=\frac{3\eta V}{kT},\qquad \varphi=nV,\qquad \tan\alpha=\omega\tau_B.\tag{39.2.1}$$

39.3 Assumptions

Those of Volume IV: weak field ($\xi\ll 1$), slow flow ($\Omega\tau_B\ll 1$), rigid dipoles, steady state, $\boldsymbol{H}\perp\boldsymbol{\Omega}$. Nothing new is added.

39.4 Step-by-Step Derivation

Step 1: only the z component survives. The field lies along x, $\boldsymbol{H}=(H_0\cos\omega t,0,0)$, and the magnetisation lies in the xy plane, $\boldsymbol{M}=(M_x,M_y,0)$. By the component definition of

the cross product (1.5.3),

$$\big(\boldsymbol{M}\times\boldsymbol{H}\big)_x = M_yH_z - M_zH_y = 0 - 0 = 0, \tag{39.4.1}$$

$$\big(\boldsymbol{M}\times\boldsymbol{H}\big)_y = M_zH_x - M_xH_z = 0 - 0 = 0, \tag{39.4.2}$$

$$\big(\boldsymbol{M}\times\boldsymbol{H}\big)_z = M_xH_y - M_yH_x = 0 - M_yH_x = -M_yH_x. \tag{39.4.3}$$

So the torque is purely axial — along the same axis as the vorticity, which is exactly what is needed if it is to speed up or slow down the rotation.

Notice what has just happened: the longitudinal magnetisation M_x, which is by far the larger of the two components, has dropped out completely. A magnetisation parallel to the field does no mechanical work. Only the small transverse piece $M_y \propto \Omega$ matters.

Step 2: insert M_y and H_x. With (SM-23) and (SM-14),

$$\big(\boldsymbol{M}\times\boldsymbol{H}\big)_z = -\Big[\Omega\tau_B\chi H_0\cos^2\alpha\cos(\omega t-2\alpha)\Big]\Big[H_0\cos\omega t\Big] \tag{39.4.4}$$

$$= -\Omega\tau_B\chi H_0^2\cos^2\alpha\,\cos\omega t\,\cos(\omega t-2\alpha), \tag{39.4.5}$$

which is the first line of (SM-25). ✓

Step 3: expand the phase-shifted cosine. By the addition formula $\cos(A-B) = \cos A\cos B + \sin A\sin B$ with $A=\omega t$, $B=2\alpha$,

$$\cos\omega t\,\cos(\omega t-2\alpha) = \cos\omega t\Big[\cos\omega t\cos 2\alpha+\sin\omega t\sin 2\alpha\Big] = \cos^2\omega t\,\cos 2\alpha+\sin\omega t\cos\omega t\,\sin 2\alpha. \tag{39.4.6}$$

The two terms have quite different characters, as the next chapter will show: the first has a nonzero average over a period, the second does not.

Step 4: convert the prefactor. We must show

$$\tau_B\chi H_0^2 = 2\eta\varphi\xi^2. \tag{39.4.7}$$

Insert the definitions:

$$\tau_B\chi H_0^2 = \frac{3\eta V}{kT}\cdot\frac{nm^2}{3kT}\cdot H_0^2 \tag{39.4.8}$$

$$= \frac{3\,\eta\,V\,n\,m^2H_0^2}{3\,(kT)^2} \qquad \text{(collecting factors)} \tag{39.4.9}$$

$$= \eta\,(nV)\,\frac{m^2H_0^2}{(kT)^2} \qquad \text{(the 3's cancel)} \tag{39.4.10}$$

$$= \eta\varphi\cdot\frac{m^2H_0^2}{(kT)^2} \qquad (\varphi = nV) \tag{39.4.11}$$

$$= \eta\varphi\cdot 2\xi^2, \qquad \text{since } \xi^2 = \frac{m^2H_0^2}{2(kT)^2} \tag{39.4.12}$$

which is (39.4.7). ✓

Step 5: assemble. Substituting (39.4.6) and (39.4.7) into (39.4.5):

$$\boldsymbol{M}\times\boldsymbol{H} = -2\Omega\eta\varphi\xi^2\cos^2\alpha\Big(\cos^2\omega t\,\cos 2\alpha+\sin\omega t\cos\omega t\,\sin 2\alpha\Big)\hat{\boldsymbol{z}}, \tag{SM-25}$$

which is the second line of (SM-25). ■

39.5 Physical Interpretation

The sign. The torque is proportional to $-\Omega\cos 2\alpha$ (once the oscillating factor is averaged, as in Chapter 41). For $\omega\tau_B < 1$ we have $\cos 2\alpha > 0$ and the torque *opposes* the vorticity — the familiar magnetoviscous braking. For $\omega\tau_B > 1$, $\cos 2\alpha < 0$ and the torque acts *with* the vorticity. Nothing in the algebra forbids this; the field is doing work, and it is allowed to give as well as to take.

The two time scales. The authors point out immediately that (SM-25) "contains two time scales: the 'fast' one and the 'slow' one". The fast time appears explicitly, in the products of $\cos\omega t$ and $\sin\omega t$; the slow one is hidden inside Ω, which is set by the hydrodynamics and changes only on the scale of the flow. Sorting out the two is the business of Chapter 40.

Order of magnitude. The torque is of order $\eta\varphi\xi^2\Omega$. Compared with the viscous torque density $6\eta\varphi\Omega$ that would be needed to stop the particles rotating altogether, it is smaller by $\sim \xi^2/6$ — the same small factor that appeared in Θ/ω in Eq. (SM-19), and that will appear once more in (SM-33).

39.6 Connection to the Next Equation

To use (SM-25) in the momentum equation we must first establish that the fluid responds only to the *average* of this rapidly oscillating torque. That is the content of (SM-26).

Chapter 40

Equation (SM-26): Separation of the Time Scales

$$\rho \frac{\mathrm{d}\boldsymbol{v}}{\mathrm{d}t} = -\nabla p + \overline{(\boldsymbol{M}\cdot\nabla)\boldsymbol{H}} + \eta\Delta\boldsymbol{v} + \tfrac{1}{2}\nabla\times\overline{\big(\boldsymbol{M}\times\boldsymbol{H}\big)}. \tag{SM-26}$$

40.1 Context

The magnetic torque (39.4.5) oscillates at 2ω (through $\cos^2\omega t$ and $\sin\omega t\cos\omega t$). The flow cannot possibly follow such rapid forcing: the liquid has inertia, and momentum diffuses only as fast as viscosity allows. Equation (SM-26) is the formal statement that what the flow feels is the *period average* of the torque, denoted by an overbar. Establishing it requires a two-scale (multiple-scale) expansion, which is a standard technique of applied mathematics but must, by our rules, be derived rather than invoked.

40.2 Symbols

Symbol	Meaning	Type	Units
$\boldsymbol{v}_0,\ p_0$	slowly varying parts of the velocity and pressure	fields	$\mathrm{cm\,s^{-1}}$, $\mathrm{dyn\,cm^{-2}}$
$\boldsymbol{v}_1,\ p_1$	small, fast-oscillating additions at frequency ω	fields	as above
$\overline{X}$	average of X over one field period $2\pi/\omega$	operator	—
$\tau_h \sim l^2/\nu$	characteristic hydrodynamic time	scalar	s
l	linear length scale of the flow (here the capillary radius)	scalar	cm
$\nu = \eta/\rho$	kinematic viscosity	scalar	$\mathrm{cm^2 s^{-1}}$

Definition 40.2.1 (Period average)**.**

$$\overline{X} \equiv \frac{\omega}{2\pi}\int_{t}^{t+2\pi/\omega} X(t')\,\mathrm{d}t'. \tag{40.2.1}$$

For a quantity that varies slowly, the average returns the quantity itself; for one that oscillates at ω or any multiple of it, the average is zero. Both facts are used below.

40.3 Assumptions

Assumption 40.3.1 (Scale separation)**.** The field period is much shorter than the hydrodynamic response time:

$$\frac{2\pi}{\omega} \ll \tau_h \sim \frac{l^2}{\nu} = \frac{\rho l^2}{\eta}. \tag{40.3.1}$$

Justification, with the experimental numbers. For Bacri *et al.*, the capillary radius is $l = 0.05$ cm, the zero-field viscosity is $\eta = 0.77$ P, and the density of a 20% suspension of ferrite in water is $\rho \approx 2\ \mathrm{g\,cm^{-3}}$. Hence

$$\tau_h = \frac{\rho l^2}{\eta} = \frac{2 \times (0.05)^2}{0.77} = \frac{5.0 \times 10^{-3}}{0.77} = 6.5 \times 10^{-3}\ \mathrm{s}. \tag{40.3.2}$$

Compare with the field period $2\pi/\omega = 1/f$:

f	$1/f$	$(1/f)/\tau_h$
52 Hz	19 ms	2.9
150 Hz	6.7 ms	1.0
645 Hz	1.6 ms	0.24
1480 Hz	0.68 ms	0.10

The separation is comfortable at the high frequencies — which is precisely where the negative-viscosity effect lives — and only marginal at the lowest ones, where the effect is positive and small anyway. It is honest to record this: the averaging procedure is a good approximation in the regime the papers care about, and a borderline one at 50 Hz. (Bacri *et al.* state the same criterion in the form $2\pi/\omega \ll \rho d^2/\eta$ before their Eq. (B-9), with d the tube diameter, which is four times more generous and comfortably satisfied throughout.)

Assumption 40.3.2 (No slow part in $\boldsymbol{M}$ and $\boldsymbol{H}$)**.** The magnetisation and field are *purely* fast: they oscillate at ω with no slowly varying component. This is evident from (SM-14) and (SM-23).

40.4 Step-by-Step Derivation

Step 1: split the hydrodynamic fields. Write

$$\boldsymbol{v} = \boldsymbol{v}_0 + \boldsymbol{v}_1, \qquad p = p_0 + p_1, \tag{40.4.1}$$

where $\boldsymbol{v}_0, p_0$ change only on the slow hydrodynamic time scale and $\boldsymbol{v}_1, p_1$ are the small responses at the field frequency. That such a split is possible, and that $\boldsymbol{v}_1 \ll \boldsymbol{v}_0$, follows from Assumption 40.3.1: a forcing of period T_0 acting on a system with response time $\tau_h \gg T_0$ produces a response smaller by the factor T_0/τ_h.

Step 2: substitute into the momentum equation. Insert (40.4.1) into (SM-12) and expand the material derivative $\mathrm{d}/\mathrm{d}t = \partial_t + (\boldsymbol{v}\cdot\nabla)$, keeping terms to first order in the small quantities $\boldsymbol{v}_1, p_1$:

$$\rho\left(\frac{\partial \boldsymbol{v}_0}{\partial t} + \frac{\partial \boldsymbol{v}_1}{\partial t} + (\boldsymbol{v}_1\cdot\nabla)\boldsymbol{v}_0\right) = -\nabla(p_0+p_1) + (\boldsymbol{M}\cdot\nabla)\boldsymbol{H} + \eta\Delta(\boldsymbol{v}_0+\boldsymbol{v}_1) + \tfrac{1}{2}\nabla\times(\boldsymbol{M}\times\boldsymbol{H}), \tag{40.4.2}$$

where now $\mathrm{d}/\mathrm{d}t = \partial/\partial t + (\boldsymbol{v}_0\cdot\nabla)$ acting on $\boldsymbol{v}_0$ has been written out and the term $(\boldsymbol{v}_1\cdot\nabla)\boldsymbol{v}_1$, being second order, dropped. This is the unnumbered equation printed by the authors immediately before their (SM-26).

Step 3: average over the fast time. Apply (40.2.1) to every term. During one field period the slow fields are effectively constant, so $\overline{\boldsymbol{v}_0} = \boldsymbol{v}_0$ and $\overline{p_0} = p_0$: they pass through the average untouched. The fast fields, on the other hand, are strictly periodic with the field period and have zero mean:

$$\overline{\boldsymbol{v}_1} = \boldsymbol{0}, \qquad \overline{p_1} = 0. \tag{40.4.3}$$

Why they have zero mean: $\boldsymbol{v}_1$ is driven linearly by the oscillating part of the torque, which by (39.4.6) contains only $\cos^2\omega t$ and $\sin\omega t\cos\omega t$; subtracting their averages leaves pure $\cos 2\omega t$ and $\sin 2\omega t$, both of zero mean by Eq. (27.4.2). A linear response to a zero-mean periodic drive is itself zero-mean.

Consequently *every term linear in* $\boldsymbol{v}_1$ *or* p_1 *averages to zero*:

$$\overline{\frac{\partial \boldsymbol{v}_1}{\partial t}} = \frac{\omega}{2\pi}\Big[\boldsymbol{v}_1\Big]_t^{t+2\pi/\omega} = \boldsymbol{0} \qquad \text{(the endpoints are equal, by periodicity)}, \tag{40.4.4}$$

$$\overline{(\boldsymbol{v}_1\cdot\nabla)\boldsymbol{v}_0} = (\overline{\boldsymbol{v}_1}\cdot\nabla)\boldsymbol{v}_0 = \boldsymbol{0} \qquad (\boldsymbol{v}_0 \text{ is constant over the average}), \tag{40.4.5}$$

$$\overline{\nabla p_1} = \nabla\overline{p_1} = 0, \qquad \overline{\eta\Delta\boldsymbol{v}_1} = \eta\Delta\overline{\boldsymbol{v}_1} = \boldsymbol{0}. \tag{40.4.6}$$

In the last two lines the spatial derivatives were interchanged with the time average, which is legitimate because the average is an integral over t at fixed $\boldsymbol{r}$.

Step 4: what survives. The magnetic terms do *not* vanish, because they are quadratic in the oscillating quantities and therefore have nonzero mean — this is the whole point. Collecting:

$$\rho\left(\frac{\partial \boldsymbol{v}_0}{\partial t} + (\boldsymbol{v}_0\cdot\nabla)\boldsymbol{v}_0\right) = -\nabla p_0 + \overline{(\boldsymbol{M}\cdot\nabla)\boldsymbol{H}} + \eta\Delta\boldsymbol{v}_0 + \tfrac{1}{2}\nabla\times\overline{(\boldsymbol{M}\times\boldsymbol{H})}. \tag{40.4.7}$$

Step 5: drop the subscript. Since the fast parts have served their purpose and been eliminated, the authors write $\boldsymbol{v}$ for $\boldsymbol{v}_0$ and p for p_0, giving

$$\rho\frac{\mathrm{d}\boldsymbol{v}}{\mathrm{d}t} = -\nabla p + \overline{(\boldsymbol{M}\cdot\nabla)\boldsymbol{H}} + \eta\Delta\boldsymbol{v} + \tfrac{1}{2}\nabla\times\overline{(\boldsymbol{M}\times\boldsymbol{H})}, \tag{SM-26}$$

which is Eq. (SM-26). ■

40.5 Physical Interpretation

The oscillating torque shakes the fluid back and forth at 2ω, but the fluid is too sluggish to move appreciably in half a period; what it does respond to is the *rectified* part — the nonzero average that survives because the torque is quadratic in oscillating quantities. This is the same mechanism by which an alternating current produces a steady heating, or a sound wave produces a steady acoustic streaming: a nonlinear response to a zero-mean drive need not have zero mean.

Remark 40.5.1 (Why $\boldsymbol{M}$ and $\boldsymbol{H}$ are *not* split). The authors point out that no such slow/fast splitting is needed for $\boldsymbol{M}$ and $\boldsymbol{H}$, since these quantities oscillate rapidly and carry no slowly varying part at all. This is Assumption 40.3.2, and it is what makes the procedure clean: the magnetic quantities are purely fast, the hydrodynamic ones are predominantly slow, and the coupling between them survives only through the averaged products.

40.6 Connection to the Next Equation

Performing the average of (SM-25) and comparing the result with the viscous term gives the rotational viscosity, (SM-27).

Chapter 41

Equation (SM-27): The Rotational Viscosity

$$\Delta\eta = \tfrac{1}{4}\eta\varphi\xi^2 \cos^2\alpha\, \cos 2\alpha. \tag{SM-27}$$

41.1 Context

We now perform the average called for by (SM-26), substitute into the momentum equation, and read off the coefficient of $\nabla^2\boldsymbol{v}$. The procedure repeats Sec. 14.4 of Volume II, but with an averaged torque in place of a static one.

41.2 Step-by-Step Derivation

Step 1: average the torque. Apply the period average to (SM-25). The slowly varying factors Ω, $\cos^2\alpha$, $\cos 2\alpha$, $\sin 2\alpha$ are constants over one period and pass outside. There remain two elementary averages, established as Eqs. (27.4.5) and (27.4.6):

$$\overline{\cos^2\omega t} = \tfrac{1}{2}, \qquad \overline{\sin\omega t\cos\omega t} = 0. \tag{41.2.1}$$

Hence

$$\overline{\boldsymbol{M}\times\boldsymbol{H}} = -2\Omega\eta\varphi\xi^2\cos^2\alpha\Big(\overline{\cos^2\omega t}\,\cos 2\alpha + \overline{\sin\omega t\cos\omega t}\,\sin 2\alpha\Big)\hat{\boldsymbol{z}} \tag{41.2.2}$$

$$= -2\Omega\eta\varphi\xi^2\cos^2\alpha\Big(\tfrac{1}{2}\cos 2\alpha + 0\Big)\hat{\boldsymbol{z}} \tag{41.2.3}$$

$$= -\Omega\eta\varphi\xi^2\cos^2\alpha\,\cos 2\alpha\,\hat{\boldsymbol{z}}. \tag{41.2.4}$$

Since $\boldsymbol{\Omega} = \Omega\hat{\boldsymbol{z}}$, this may be written as a vector relation:

$$\boxed{\overline{\boldsymbol{M}\times\boldsymbol{H}} = -\eta\varphi\xi^2\cos^2\alpha\cos 2\alpha\,\boldsymbol{\Omega}} \tag{41.2.5}$$

This is the unnumbered result the authors record when they note that averaging (SM-25) leaves $\tfrac{1}{2}\cos 2\alpha$ from the first bracketed term and nothing from the second. ✓

Step 2: the disappearance of the second term, examined. It is worth pausing on why $\overline{\sin\omega t\cos\omega t} = 0$ kills the $\sin 2\alpha$ contribution. That term represents the part of the transverse magnetisation that is *in quadrature* with the field. A torque in quadrature with the driving

field does work over the first half of a cycle and takes it back over the second: it transports no net momentum to the fluid. Only the in-phase part, carried by $\cos 2\alpha$, survives averaging. This is the same distinction as between the reactive and resistive parts of an impedance.

Step 3: insert into the momentum equation. By (SM-26) the magnetic contribution to the force density is $\frac{1}{2}\nabla \times \overline{\boldsymbol{M} \times \boldsymbol{H}}$. With (41.2.5),

$$\tfrac{1}{2}\nabla \times \overline{\boldsymbol{M} \times \boldsymbol{H}} = -\tfrac{1}{2}\eta\varphi\xi^2 \cos^2\alpha \cos 2\alpha \, \nabla \times \boldsymbol{\Omega}. \tag{41.2.6}$$

Step 4: express the viscous term the same way. By the definition $\boldsymbol{\Omega} = \frac{1}{2}\nabla \times \boldsymbol{v}$ (3.4.1) and the identity (2.6.1) together with incompressibility (5.1.4),

$$\eta\Delta\boldsymbol{v} = \eta\nabla^2\boldsymbol{v} = -\eta\,\nabla \times \big(\nabla \times \boldsymbol{v}\big) = -2\eta\,\nabla \times \boldsymbol{\Omega}. \tag{41.2.7}$$

Step 5: add them. Summing (41.2.7) and (41.2.6):

$$\eta\Delta\boldsymbol{v} + \tfrac{1}{2}\nabla \times \overline{\boldsymbol{M} \times \boldsymbol{H}} = -2\eta\nabla \times \boldsymbol{\Omega} - \tfrac{1}{2}\eta\varphi\xi^2 \cos^2\alpha \cos 2\alpha \, \nabla \times \boldsymbol{\Omega} \tag{41.2.8}$$

$$= -2\Big(\eta + \tfrac{1}{4}\eta\varphi\xi^2 \cos^2\alpha \cos 2\alpha\Big)\nabla \times \boldsymbol{\Omega}, \tag{41.2.9}$$

where in the last line the factor 2 was extracted: $\frac{1}{2} = 2 \times \frac{1}{4}$. This is precisely the line printed in the paper,

$$\eta\Delta\boldsymbol{v} - \tfrac{1}{2}\eta\varphi\xi^2 \cos^2\alpha \cos 2\alpha \, \nabla \times \boldsymbol{\Omega} = -2\Big(\eta + \tfrac{1}{4}\eta\varphi\xi^2 \cos^2\alpha \cos 2\alpha\Big)\nabla \times \boldsymbol{\Omega}.$$

Step 6: read off the viscosity. Comparison of (41.2.9) with the field-free case, where the same combination is $-2\eta\nabla \times \boldsymbol{\Omega}$, shows that the fluid behaves exactly as a Newtonian liquid of viscosity $\eta + \Delta\eta$ with

$$\boxed{\Delta\eta = \tfrac{1}{4}\eta\varphi\xi^2 \cos^2\alpha \, \cos 2\alpha} \tag{SM-27}$$

■

41.3 Consistency Check Against the Master Formula

Volume II established the general relation $\Delta\eta = \frac{3}{2}\eta\varphi(\Omega - \Theta)/\Omega$, Eq. (14.4.9). Using (41.2.5) together with the torque balance $\overline{\boldsymbol{M} \times \boldsymbol{H}} = -6\eta\varphi(\boldsymbol{\Omega} - \boldsymbol{\Theta})$, Eq. (14.2.1):

$$\boldsymbol{\Omega} - \boldsymbol{\Theta} = \frac{\eta\varphi\xi^2 \cos^2\alpha \cos 2\alpha}{6\eta\varphi}\boldsymbol{\Omega} = \frac{\xi^2}{6}\cos^2\alpha \cos 2\alpha \, \boldsymbol{\Omega}, \tag{41.3.1}$$

so that

$$\Delta\eta = \frac{3}{2}\eta\varphi \cdot \frac{\xi^2}{6}\cos^2\alpha \cos 2\alpha = \frac{\eta\varphi\xi^2}{4}\cos^2\alpha \cos 2\alpha. \quad \checkmark \tag{41.3.2}$$

The two independent routes agree exactly. Equation (41.3.1) will be used again in Chapter 47 to obtain (SM-33).

41.4 Connection to the Next Equation

Expressing $\cos^2\alpha$ and $\cos 2\alpha$ through $\omega\tau_B$ turns (SM-27) into the central result of the paper, (SM-28).

Chapter 42

Equation (SM-28): The Central Result

$$\Delta\eta = \frac{1}{4}\,\eta\varphi\,\xi^2\,\frac{1-\omega^2\tau_B^2}{\left(1+\omega^2\tau_B^2\right)^2}. \tag{SM-28}$$

42.1 Context

This is the equation for which the paper was written. Everything before it is preparation and everything after it is generalisation. It says that the rotational viscosity of a ferrofluid in a weak alternating field is positive at low frequency, passes through zero at $\omega\tau_B = 1$, and is *negative* above it.

42.2 Step-by-Step Derivation

Step 1. From Eq. (33.4.1) with $x = \tan\alpha = \omega\tau_B$,

$$\cos^2\alpha = \frac{1}{1+\omega^2\tau_B^2}. \tag{42.2.1}$$

Step 2. From Eq. (37.5.8),

$$\cos 2\alpha = \frac{1-\tan^2\alpha}{1+\tan^2\alpha} = \frac{1-\omega^2\tau_B^2}{1+\omega^2\tau_B^2}. \tag{42.2.2}$$

Step 3. Multiply and substitute into (SM-27):

$$\Delta\eta = \frac{1}{4}\eta\varphi\xi^2 \cdot \frac{1}{1+\omega^2\tau_B^2} \cdot \frac{1-\omega^2\tau_B^2}{1+\omega^2\tau_B^2} \tag{42.2.3}$$

$$= \frac{1}{4}\eta\varphi\xi^2\,\frac{1-\omega^2\tau_B^2}{\left(1+\omega^2\tau_B^2\right)^2}. \tag{SM-28}$$

■

42.3 Analysis of the Function

Write $y \equiv \omega\tau_B$ and define the relative rotational viscosity plotted in the paper's Fig. 1,

$$R(y) \equiv \frac{4\Delta\eta}{\eta\varphi\xi^2} = \frac{1-y^2}{\left(1+y^2\right)^2}. \tag{42.3.1}$$

(a) Value at zero frequency. $R(0) = 1$, so

$$\Delta\eta(0) = \tfrac{1}{4}\eta\varphi\xi^2, \tag{42.3.2}$$

which is formally identical to the static formula (SM-3). The authors say (SM-28) "provides the correct value of $\Delta\eta$ at $\omega = 0$ [see (3)]". But there is a subtlety worth making explicit, because it is the deepest justification of the "acting field" convention of Sec. 33.2.

At $\omega = 0$ the field is not static — it is $H_0 \cos\omega t$ with ω merely very small. The instantaneous viscosity is then the static formula (SM-3) evaluated at the instantaneous field, and what a slow measurement returns is its average over the slow cycle:

$$\overline{\Delta\eta_{\text{stat}}} = \overline{\frac{\eta\varphi}{4}\left(\frac{mH_0\cos\omega t}{kT}\right)^2} \tag{42.3.3}$$

$$= \frac{\eta\varphi}{4}\frac{m^2H_0^2}{(kT)^2}\,\overline{\cos^2\omega t} \tag{42.3.4}$$

$$= \frac{\eta\varphi}{4}\cdot\frac{m^2H_0^2}{(kT)^2}\cdot\frac{1}{2} \tag{42.3.5}$$

$$= \frac{\eta\varphi}{4}\cdot\xi^2, \qquad \text{by (39.4.7)'s definition } \xi^2 = \frac{m^2H_0^2}{2(kT)^2} \tag{42.3.6}$$

exactly reproducing (42.3.2). *This is why ξ was defined over $H_0/\sqrt{2}$*: with that convention, and no other, the $\omega \to 0$ limit of the alternating-field theory agrees with the cycle-average of the static theory without stray factors of two. ✓

(b) The zero crossing. $R(y) = 0$ requires $1 - y^2 = 0$, i.e.

$$\boxed{\omega_0\tau_B = 1} \tag{42.3.7}$$

Below this frequency the ferrofluid is thickened by the field; above it, thinned below its zero-field value. The condition has a transparent meaning: it is the frequency at which the magnetisation lags the field by exactly $\alpha = \pi/4$, so that the transverse magnetisation, which lags by $2\alpha = \pi/2$, is exactly in quadrature with the field and delivers no average torque at all.

(c) The minimum. Differentiate (42.3.1) using the quotient rule, with $u = 1-y^2$ ($u' = -2y$) and $v = (1+y^2)^2$ ($v' = 2(1+y^2)\cdot 2y = 4y(1+y^2)$):

$$R'(y) = \frac{-2y\left(1+y^2\right)^2 - \left(1-y^2\right)\cdot 4y\left(1+y^2\right)}{\left(1+y^2\right)^4} \tag{42.3.8}$$

$$= \frac{\left(1+y^2\right)\Big[-2y\left(1+y^2\right) - 4y\left(1-y^2\right)\Big]}{\left(1+y^2\right)^4} \qquad \text{(factor out } (1+y^2)\text{)} \tag{42.3.9}$$

$$= \frac{-2y\big(1+y^2\big) - 4y\big(1-y^2\big)}{\big(1+y^2\big)^3} \tag{42.3.10}$$

$$= \frac{-2y - 2y^3 - 4y + 4y^3}{\big(1+y^2\big)^3} \tag{42.3.11}$$

$$= \frac{2y\big(y^2-3\big)}{\big(1+y^2\big)^3}. \tag{42.3.12}$$

This vanishes at $y = 0$ (the maximum, $R = 1$) and at $y^2 = 3$:

$$\boxed{\omega_m \tau_B = \sqrt{3} \approx 1.732} \tag{42.3.13}$$

That it is a minimum follows from the sign of R': for $0 < y < \sqrt{3}$ the factor $(y^2 - 3)$ is negative so R decreases; for $y > \sqrt{3}$ it is positive so R increases.

(d) The depth of the minimum. Evaluate (42.3.1) at $y = \sqrt{3}$:

$$R(\sqrt{3}) = \frac{1-3}{\big(1+3\big)^2} = \frac{-2}{16} = -\frac{1}{8}, \tag{42.3.14}$$

so that

$$\boxed{\Delta\eta\big(\omega_m\big) = -\frac{\Delta\eta(0)}{8}} \tag{42.3.15}$$

exactly as the authors state ("at this point $\Delta\eta(\sqrt{3}) = -\Delta\eta(0)/8$"). The negative excursion is thus *eight times smaller* than the positive low-frequency value — a quantitative fact that matters for experiment: the effect is real but modest, and Bacri *et al.* indeed measure a viscosity reduction of about 25% rather than a dramatic collapse.

(e) The large-frequency tail. For $y \gg 1$,

$$R(y) \approx \frac{-y^2}{y^4} = -\frac{1}{y^2}, \tag{42.3.16}$$

so $\Delta\eta$ returns to zero from below as ω^{-2}. This is (SM-29), the subject of the next chapter.

(f) Summary of the shape. R starts at 1, falls monotonically, crosses zero at $y = 1$, reaches $-1/8$ at $y = \sqrt{3}$, and climbs back towards zero as $-1/y^2$. This is exactly the curve of the paper's Fig. 1.

42.4 Physical Interpretation: What Is Actually Happening

The chain of causation, assembled from all of Volumes IV and V, runs as follows.

1. The shear tries to rotate the grains at Ω.
2. Rotating a grain shifts the frequency at which it sees each of the two circular components of the field — to $\omega - \Omega$ for the co-rotating one and $\omega + \Omega$ for the counter-rotating one (Sec. 35.4).

3. Because the Debye response is a nonlinear function of frequency, the two components no longer produce equal and opposite torques. The imbalance is a net transverse magnetisation $M_y \propto \Omega$, lagging the field by 2α (Sec. 37.5).

4. That transverse magnetisation, crossed with the field, gives a torque whose average is $\propto -\Omega \cos 2\alpha$ (Sec. 41.2).

5. At low frequency $\cos 2\alpha > 0$: the torque opposes the rotation, the grains turn slower than the fluid, the liquid must shear past them, and the viscosity rises.

6. At high frequency $\cos 2\alpha < 0$: the torque *drives* the rotation, the grains turn *faster* than the fluid, they hand angular momentum to it, and the viscosity falls.

Where does the energy come from? From the field. Bacri *et al.* summarise the route in their abstract: energy stored in the field is converted first into angular momentum of the grains, and then by the grains into hydrodynamic motion of the liquid. There is no violation of anything: the alternating field is maintained by a generator doing work on the solenoid, and a fraction of that work ends up as kinetic energy of the flow. Contrast the static case, where a permanent magnet — which does no work — could produce the field, and where $\Delta\eta$ is therefore required by thermodynamics to be positive (Sec. 15.6).

Why the effect must vanish at both ends. At $\omega \to 0$ the field is quasi-static and the thermodynamic argument applies, forcing $\Delta\eta > 0$. At $\omega \to \infty$ the moments cannot follow at all, $M \to 0$, and there is no torque of either sign. The negative region is squeezed between these two limits and is therefore necessarily bounded — hence the modest factor $1/8$ of (42.3.15).

42.5 Connection to the Next Equation

The paper next records the explicit high-frequency asymptote, (SM-29).

Chapter 43

Equation (SM-29): The High-Frequency Asymptote

$$\Delta\eta = -\frac{\varphi}{\eta}\left(\frac{M_s H}{6\omega}\right)^2 \qquad \text{at} \quad \omega\tau_B \gg 1. \tag{SM-29}$$

43.1 Context

The authors record this limit because it shows something initially surprising: although the viscosity tends to zero at high frequency, *the way* it does so is independent of τ_B — the Brownian time cancels out. The high-frequency ferrofluid has forgotten how big its particles are.

43.2 Symbols

M_s is the spontaneous magnetisation of the grain material (G); H is the acting field $H_0/\sqrt{2}$, consistently with the convention of Sec. 33.2 — as the derivation below confirms.

43.3 Step-by-Step Derivation

Step 1: take the limit of (SM-28). For $\omega\tau_B \gg 1$ the numerator $1-\omega^2\tau_B^2 \to -\omega^2\tau_B^2$ and the denominator $(1+\omega^2\tau_B^2)^2 \to \omega^4\tau_B^4$, so

$$\Delta\eta \to \frac{\eta\varphi\xi^2}{4}\cdot\frac{-\omega^2\tau_B^2}{\omega^4\tau_B^4} = -\frac{\eta\varphi\xi^2}{4\,\omega^2\tau_B^2}. \tag{43.3.1}$$

Step 2: evaluate ξ^2/τ_B^2. Insert $\xi^2 = m^2H_0^2/2(kT)^2$ and $\tau_B = 3\eta V/kT$:

$$\frac{\xi^2}{\tau_B^2} = \frac{m^2H_0^2}{2(kT)^2}\cdot\frac{(kT)^2}{9\eta^2V^2} \tag{43.3.2}$$

$$= \frac{m^2H_0^2}{18\,\eta^2V^2} \qquad ((kT)^2 \text{ cancels}) \tag{43.3.3}$$

$$= \frac{\left(M_sV\right)^2H_0^2}{18\,\eta^2V^2} \qquad \text{using } m = M_sV \tag{43.3.4}$$

$$= \frac{M_s^2 H_0^2}{18\,\eta^2}. \qquad (V^2 \text{ cancels}) \tag{43.3.5}$$

The volume has disappeared — and with it the particle size, and with it τ_B.

Step 3: assemble. Substituting (43.3.5) into (43.3.1):

$$\Delta\eta = -\frac{\eta\varphi}{4\omega^2} \cdot \frac{M_s^2 H_0^2}{18\eta^2} \tag{43.3.6}$$

$$= -\frac{\varphi M_s^2 H_0^2}{72\,\eta\,\omega^2}. \tag{43.3.7}$$

Step 4: recognise the paper's form. Write the acting field $H = H_0/\sqrt{2}$, so $H_0^2 = 2H^2$:

$$\Delta\eta = -\frac{\varphi M_s^2 \cdot 2H^2}{72\,\eta\,\omega^2} = -\frac{\varphi M_s^2 H^2}{36\,\eta\,\omega^2} = -\frac{\varphi}{\eta} \cdot \frac{M_s^2 H^2}{36\,\omega^2} = -\frac{\varphi}{\eta}\left(\frac{M_s H}{6\omega}\right)^2, \tag{SM-29}$$

since $36 = 6^2$. ■

43.4 Physical Interpretation

Why τ_B drops out. At high frequency the magnetisation amplitude is $M \simeq \chi H_0/\omega\tau_B$, which falls with τ_B; but the torque per unit magnetisation grows with the lag, which rises with τ_B. The two dependences cancel. What remains is a competition between the magnetic energy density $M_s H$ and the frequency, moderated by the viscosity.

The sign and the scaling. $\Delta\eta < 0$ always in this regime, and $|\Delta\eta| \propto \omega^{-2}$. Doubling the frequency quarters the effect. Since $\Delta\eta$ must also vanish as $\omega \to 0$, there is necessarily an optimum, and Eq. (42.3.13) located it at $\omega\tau_B = \sqrt{3}$.

An experimental corollary. Because (43.3.7) contains η in the *denominator*, the negative viscosity at fixed high frequency is larger for a *less* viscous carrier. Together with the φ and M_s^2 dependences this is the design rule for a fluid in which the effect is large: high magnetisation, high concentration, low carrier viscosity — and, from (42.3.13), a particle size chosen so that τ_B places $\sqrt{3}/\tau_B$ in an accessible frequency range. Bacri *et al.* made exactly these choices.

43.5 Connection to the Next Equation

Having completed the first variant of the solution, the authors present the second: a direct calculation in the linearly polarised field, without any decomposition. That begins at (SM-30).

Chapter 44

Equation (SM-30): The Second Variant — Longitudinal Magnetisation

$$M_x = \chi H_0 \cos\alpha \, \cos(\omega t - \alpha), \qquad \tan\alpha = \omega\tau_B. \tag{SM-30}$$

44.1 Context

The authors remark that the same Eq. (SM-28) can be reached more quickly by working directly with the linearly polarised field (SM-14). The second variant is quicker but less transparent; the authors give both because each illuminates something the other hides. The first showed *why* the effect exists (the Doppler imbalance between two circular components); the second shows how little machinery is really needed.

We present it in full, both because completeness demands it and because it provides an independent check on the long calculation of Volume IV.

44.2 Assumptions

Same as before. In this chapter the fluid is at rest, $\mathbf{\Omega} = 0$; the vortex enters in Chapter 45.

44.3 Mathematical Background: forced linear relaxation

We shall need the steady periodic solution of

$$\tau_B \frac{\mathrm{d}y}{\mathrm{d}t} + y = A\cos\omega t. \tag{44.3.1}$$

Result 44.3.1. *The steady solution is*

$$y(t) = A\cos\alpha \, \cos(\omega t - \alpha), \qquad \tan\alpha = \omega\tau_B. \tag{44.3.2}$$

Derivation. Try $y = B\cos(\omega t - \alpha)$ with B and α to be determined. Then $\mathrm{d}y/\mathrm{d}t = -B\omega\sin(\omega t - \alpha)$ and the left-hand side of (44.3.1) is

$$B\Big[\cos(\omega t - \alpha) - \omega\tau_B \sin(\omega t - \alpha)\Big]. \tag{44.3.3}$$

To combine the two terms into a single cosine, use the identity

$$\cos u - \lambda \sin u = \sqrt{1+\lambda^2}\,\cos(u+\alpha) \qquad \text{with} \qquad \tan\alpha = \lambda, \tag{44.3.4}$$

which is proved by expanding the right-hand side: $\sqrt{1+\lambda^2}[\cos u\cos\alpha - \sin u\sin\alpha]$, and noting from Eq. (33.4.1) that $\sqrt{1+\lambda^2}\cos\alpha = 1$ and $\sqrt{1+\lambda^2}\sin\alpha = \lambda$. Applying (44.3.4) to (44.3.3) with $u = \omega t - \alpha$ and $\lambda = \omega\tau_B = \tan\alpha$ — the *same* α, which is the trick —

$$B\sqrt{1+\omega^2\tau_B^2}\,\cos\big(\omega t - \alpha + \alpha\big) = B\sqrt{1+\omega^2\tau_B^2}\,\cos\omega t. \tag{44.3.5}$$

Matching to the right-hand side of (44.3.1),

$$B\sqrt{1+\omega^2\tau_B^2} = A \qquad \Longrightarrow \qquad B = \frac{A}{\sqrt{1+\omega^2\tau_B^2}} = A\cos\alpha, \tag{44.3.6}$$

using Eq. (33.4.1) once more. This is (44.3.2). ■

44.4 Step-by-Step Derivation of (SM-30)

Step 1: the equation for M_x in a quiescent fluid. With $\boldsymbol{\Omega} = 0$ and (as the authors note) $\boldsymbol{\Theta} = 0$ — because in the linearly polarised field the mean particle spin vanishes by symmetry, Sec. 27.5 — Eq. (SM-7) reduces to

$$\frac{\mathrm{d}\boldsymbol{M}}{\mathrm{d}t} = -\frac{1}{\tau_B}\big(\boldsymbol{M} - \boldsymbol{M}_0\big). \tag{44.4.1}$$

With $\boldsymbol{M}_0 = \chi\boldsymbol{H} = \chi H_0\cos\omega t\,\hat{\boldsymbol{x}}$ from (SM-15) and (SM-14), only the x component is driven:

$$\tau_B\frac{\mathrm{d}M_x}{\mathrm{d}t} + M_x = \chi H_0\cos\omega t. \tag{44.4.2}$$

Step 2: solve. This is (44.3.1) with $y = M_x$ and $A = \chi H_0$. By (44.3.2),

$$M_x = \chi H_0\cos\alpha\,\cos(\omega t - \alpha), \qquad \tan\alpha = \omega\tau_B, \tag{SM-30}$$

which is (SM-30). ■

44.5 Consistency with the First Variant

The first variant gave, for each circular component, $M_\pm = \chi H_0\cos\alpha$ with the same $\tan\alpha = \omega\tau_B$. Superposing the x components,

$$M_x = \tfrac{1}{2}\Big[M_+\cos(\omega t - \alpha) + M_-\cos(\omega t - \alpha)\Big] = \chi H_0\cos\alpha\cos(\omega t - \alpha), \tag{44.5.1}$$

identical to (SM-30). ✓ The two routes agree, as they must.

44.6 Physical Interpretation

Equation (SM-30) is the Debye response once again: amplitude reduced by $\cos\alpha$, phase retarded by α. The authors immediately draw the consequence: both the magnetic torque and the mean particle spin vanish. Indeed with $\boldsymbol{M} = M_x\hat{\boldsymbol{x}}$ and $\boldsymbol{H} = H_x\hat{\boldsymbol{x}}$ parallel, the cross product is identically zero. *In a quiescent fluid, a linearly polarised field does nothing at all mechanically*, however strong and however fast — the point already made qualitatively in Sec. 27.5 and now proved.

44.7 Connection to the Next Equation

Switching on the vorticity generates a y component, whose amplitude and phase obey (SM-31).

Chapter 45

Equation (SM-31): The Coupled Equations for the Transverse Component

$$\begin{aligned} a\big(\sin\theta - \omega\tau_B\cos\theta\big) &= \Omega\tau_B\chi H_0 \sin\alpha\cos\alpha, \\ a\big(\cos\theta + \omega\tau_B\sin\theta\big) &= \Omega\tau_B\chi H_0 \cos^2\alpha. \end{aligned} \tag{SM-31}$$

45.1 Context

The authors recall that a vortical flow destroys the symmetry of the problem: with $\boldsymbol{\Omega} = (0, 0, \Omega)$ a y component of the magnetisation appears, together with a z component of the particle spin. Substituting the trial form $M_y = a\cos(\omega t - \theta)$ into (SM-7) and (SM-8) then yields a coupled pair of equations for the amplitude and phase of that component.

45.2 Symbols

$a > 0$ is the amplitude of the transverse magnetisation (G) and θ its phase lag behind the field (rad). Both are unknowns to be determined; the answer, $\theta = 2\alpha$ and $a = \Omega\tau_B\chi H_0\cos^2\alpha$, is (SM-32).

45.3 Assumptions

1. Linear response in Ω: $a = O(\Omega)$, so any term containing a product of a with another small quantity is dropped.

2. The particle spin is $\boldsymbol{\Theta} \simeq \boldsymbol{\Omega}$ to the accuracy required. The magnetic correction to $\boldsymbol{\Theta}$ in (SM-11) is $O(\xi^2)$ and would contribute at order ξ^4 to the final viscosity — the same argument as in Sec. 35.3.

45.4 Step-by-Step Derivation

Step 1: write out the y component of the magnetisation equation. Equation (SM-7) with $\boldsymbol{\Theta} = \boldsymbol{\Omega} = \Omega\hat{\boldsymbol{z}}$ reads

$$\frac{\mathrm{d}\boldsymbol{M}}{\mathrm{d}t} = \boldsymbol{\Omega} \times \boldsymbol{M} - \frac{1}{\tau_B}\big(\boldsymbol{M} - \boldsymbol{M}_0\big). \tag{45.4.1}$$

Compute the convective term for $\boldsymbol{M} = (M_x, M_y, 0)$:

$$\boldsymbol{\Omega} \times \boldsymbol{M} = \Omega\hat{\boldsymbol{z}} \times \big(M_x, M_y, 0\big) = \Omega\big(-M_y,\ M_x,\ 0\big), \tag{45.4.2}$$

using $\hat{\boldsymbol{z}} \times \hat{\boldsymbol{x}} = \hat{\boldsymbol{y}}$ and $\hat{\boldsymbol{z}} \times \hat{\boldsymbol{y}} = -\hat{\boldsymbol{x}}$. Since $\boldsymbol{M}_0$ has no y component, the y equation is

$$\frac{\mathrm{d}M_y}{\mathrm{d}t} = \Omega M_x - \frac{M_y}{\tau_B}. \tag{45.4.3}$$

(The x equation is $\mathrm{d}M_x/\mathrm{d}t = -\Omega M_y - (M_x - \chi H_0 \cos\omega t)/\tau_B$; its extra term ΩM_y is $O(\Omega^2)$ and is dropped, leaving (SM-30) intact.)

Step 2: substitute the ansatz. Put $M_y = a\cos(\omega t - \theta)$ and M_x from (SM-30) into (45.4.3), and multiply throughout by τ_B:

$$-a\omega\tau_B \sin(\omega t - \theta) = \Omega\tau_B\chi H_0 \cos\alpha \cos(\omega t - \alpha) - a\cos(\omega t - \theta), \tag{45.4.4}$$

which we rearrange as

$$a\cos(\omega t - \theta) - a\omega\tau_B \sin(\omega t - \theta) = \Omega\tau_B\chi H_0 \cos\alpha\ \cos(\omega t - \alpha). \tag{45.4.5}$$

Step 3: expand both sides in $\cos\omega t$ and $\sin\omega t$. Using the addition formulas $\cos(\omega t - \theta) = \cos\omega t\cos\theta + \sin\omega t\sin\theta$ and $\sin(\omega t - \theta) = \sin\omega t\cos\theta - \cos\omega t\sin\theta$, the left-hand side becomes

$$\begin{aligned} &a\big[\cos\omega t\cos\theta + \sin\omega t\sin\theta\big] - a\omega\tau_B\big[\sin\omega t\cos\theta - \cos\omega t\sin\theta\big] \\ &\quad = \cos\omega t \cdot a\big[\cos\theta + \omega\tau_B\sin\theta\big] + \sin\omega t \cdot a\big[\sin\theta - \omega\tau_B\cos\theta\big]. \end{aligned} \tag{45.4.6}$$

The right-hand side, with $\cos(\omega t - \alpha) = \cos\omega t\cos\alpha + \sin\omega t\sin\alpha$, becomes

$$\cos\omega t \cdot \Omega\tau_B\chi H_0\cos^2\alpha + \sin\omega t \cdot \Omega\tau_B\chi H_0\cos\alpha\sin\alpha. \tag{45.4.7}$$

Step 4: match coefficients. The functions $\cos\omega t$ and $\sin\omega t$ are linearly independent — no nontrivial linear combination of them vanishes for all t, as one sees by evaluating at $t = 0$ and $\omega t = \pi/2$. Hence their coefficients in (45.4.6) and (45.4.7) must match separately:

$$\cos\omega t: \qquad a\big(\cos\theta + \omega\tau_B\sin\theta\big) = \Omega\tau_B\chi H_0\cos^2\alpha, \tag{45.4.8}$$

$$\sin\omega t: \qquad a\big(\sin\theta - \omega\tau_B\cos\theta\big) = \Omega\tau_B\chi H_0\sin\alpha\cos\alpha. \tag{45.4.9}$$

These are exactly the two lines of (SM-31). ■

45.5 Physical Interpretation

The structure is that of a driven, damped oscillator resolved into in-phase and quadrature parts. The *driving* of the transverse component, on the right, is the longitudinal magnetisation being swept sideways by the vortex — the term ΩM_x in (45.4.3). The *response*, on the left, is the transverse magnetisation fighting its own relaxation. Note that the drive is proportional to Ω and to M_x: the vortex alone would do nothing without a magnetisation to sweep, and a magnetisation alone would do nothing without a vortex.

45.6 Connection to the Next Equation

Solving the pair for a and θ gives (SM-32).

Chapter 46

Equation (SM-32): Solving the Pair

$$\theta = 2\alpha, \qquad a = \Omega\tau_B \chi H_0 \cos^2\alpha. \tag{SM-32}$$

46.1 Step-by-Step Derivation

Step 1: divide one equation by the other. Dividing (45.4.9) by (45.4.8) eliminates both a and the common factor $\Omega\tau_B\chi H_0$:

$$\frac{\sin\theta - \omega\tau_B\cos\theta}{\cos\theta + \omega\tau_B\sin\theta} = \frac{\sin\alpha\cos\alpha}{\cos^2\alpha} = \frac{\sin\alpha}{\cos\alpha} = \tan\alpha = \omega\tau_B, \tag{46.1.1}$$

the last step by (SM-24). The division is legitimate provided $a \neq 0$ and the denominator does not vanish; both hold for $0 \le \alpha < \pi/2$.

Step 2: cross-multiply. Write $\lambda \equiv \omega\tau_B$ for brevity. From (46.1.1),

$$\sin\theta - \lambda\cos\theta = \lambda\big(\cos\theta + \lambda\sin\theta\big) \tag{46.1.2}$$

$$\sin\theta - \lambda\cos\theta = \lambda\cos\theta + \lambda^2\sin\theta \tag{46.1.3}$$

$$\sin\theta - \lambda^2\sin\theta = \lambda\cos\theta + \lambda\cos\theta \tag{46.1.4}$$

$$\sin\theta\big(1 - \lambda^2\big) = 2\lambda\cos\theta. \tag{46.1.5}$$

Step 3: identify the double angle. Divide both sides by $\cos\theta(1-\lambda^2)$:

$$\tan\theta = \frac{2\lambda}{1-\lambda^2} = \frac{2\tan\alpha}{1-\tan^2\alpha} = \tan 2\alpha, \tag{46.1.6}$$

where the last equality is the tangent double-angle formula, itself derived from (46.1.8) below. Hence

$$\boxed{\theta = 2\alpha} \tag{46.1.7}$$

(the solution in the principal range; θ and 2α can differ by a multiple of π, which is fixed by requiring $a > 0$ in Step 5).

Step 4: a needed identity. For completeness,

$$\tan 2\alpha = \frac{\sin 2\alpha}{\cos 2\alpha} = \frac{2\sin\alpha\cos\alpha}{\cos^2\alpha - \sin^2\alpha} = \frac{2\tan\alpha}{1-\tan^2\alpha}, \tag{46.1.8}$$

obtained by dividing numerator and denominator by $\cos^2\alpha$.

Step 5: find the amplitude. Substitute $\theta = 2\alpha$ into (45.4.8):

$$a\big(\cos 2\alpha + \tan\alpha\,\sin 2\alpha\big) = \Omega\tau_B\chi H_0\cos^2\alpha. \tag{46.1.9}$$

The bracket collapses beautifully:

$$\begin{aligned}\cos 2\alpha + \tan\alpha\sin 2\alpha &= \cos 2\alpha + \frac{\sin\alpha}{\cos\alpha}\cdot 2\sin\alpha\cos\alpha && (46.1.10)\\ &= \cos 2\alpha + 2\sin^2\alpha && (46.1.11)\\ &= \big(\cos^2\alpha - \sin^2\alpha\big) + 2\sin^2\alpha && (46.1.12)\\ &= \cos^2\alpha + \sin^2\alpha && (46.1.13)\\ &= 1. && (46.1.14)\end{aligned}$$

Therefore (46.1.9) reduces to

$$\boxed{a = \Omega\tau_B\chi H_0\cos^2\alpha} \tag{46.1.15}$$

which completes (SM-32). ■

46.2 The Two Variants Agree

Assembling,

$$M_y = a\cos(\omega t - \theta) = \Omega\tau_B\chi H_0\cos^2\alpha\,\cos(\omega t - 2\alpha), \tag{46.2.1}$$

which is identical to (SM-23), obtained in Volume IV by an entirely different route. The authors note the coincidence with their Eq. (SM-23) and conclude that the two solution procedures give the same $\Delta\eta(\xi, \omega\tau_B)$.

Remark 46.2.1 (Why the check matters). The first variant involved a decomposition, two separate solutions, a Doppler shift, a Taylor expansion in $\Omega\tau_B$ and a reassembly — many opportunities for a lost factor. The second involved one linear ODE and a 2×2 algebraic system. That they agree exactly, term for term and phase for phase, is strong evidence that neither contains an error. Independent derivation is the only real proof-reading available in physics.

46.3 Connection to the Next Equation

The paper concludes Sec. II by computing the mean particle angular velocity, which displays the mechanism in its most direct form. That is (SM-33).

Chapter 47

Equation (SM-33): The Mean Particle Spin

$$\Theta = \Omega \left(1 - \frac{\xi^2}{6} \frac{1 - \omega^2 \tau_B^2}{\left(1 + \omega^2 \tau_B^2\right)^2} \right). \tag{SM-33}$$

47.1 Context

The authors close their Sec. II by averaging Eq. (SM-11) over the fast time scale and combining it with (SM-26) to obtain (SM-33), which they describe as a striking illustration of how the field controls the viscosity. It is indeed the most vivid statement of the result: it says directly whether the grains turn faster or slower than the fluid.

Remark 47.1.1 (A typographical note). The equation as printed in the paper reads $\Theta = \Omega\big[1 - \frac{\xi^2}{6}(1 - \xi^2\tau_B^2)/(1 + \omega^2\tau_B^2)^2\big]$. The $\xi^2\tau_B^2$ in the numerator is a misprint for $\omega^2\tau_B^2$: dimensionally ξ is a pure number so $\xi^2\tau_B^2$ would carry units of s^2, and the derivation below — as well as consistency with (SM-28) and (SM-34) — gives $\omega^2\tau_B^2$. We use the corrected form throughout.

47.2 Step-by-Step Derivation

Step 1: average (SM-11). Taking the period average of $\boldsymbol{\Theta} = \boldsymbol{\Omega} + (\tau_s/I)(\boldsymbol{M} \times \boldsymbol{H})$, and noting that $\boldsymbol{\Omega}$ is a slow quantity so $\overline{\boldsymbol{\Omega}} = \boldsymbol{\Omega}$:

$$\overline{\boldsymbol{\Theta}} = \boldsymbol{\Omega} + \frac{\tau_s}{I}\,\overline{\boldsymbol{M} \times \boldsymbol{H}} = \boldsymbol{\Omega} + \frac{1}{6\eta\varphi}\overline{\boldsymbol{M} \times \boldsymbol{H}}, \tag{47.2.1}$$

using $\tau_s/I = 1/6\eta\varphi$ from Eq. (7.5.12). From here on we write $\boldsymbol{\Theta}$ for $\overline{\boldsymbol{\Theta}}$, as the paper does.

Step 2: insert the averaged torque. From (41.2.5),

$$\boldsymbol{\Theta} = \boldsymbol{\Omega} + \frac{1}{6\eta\varphi}\Big(- \eta\varphi\xi^2 \cos^2\alpha \cos 2\alpha\, \boldsymbol{\Omega}\Big) \tag{47.2.2}$$

$$= \boldsymbol{\Omega}\left(1 - \frac{\xi^2}{6}\cos^2\alpha\cos 2\alpha\right). \tag{47.2.3}$$

Step 3: express through $\omega\tau_B$. By (42.2.1) and (42.2.2),

$$\cos^2\alpha\cos 2\alpha = \frac{1-\omega^2\tau_B^2}{\left(1+\omega^2\tau_B^2\right)^2}, \tag{47.2.4}$$

so that

$$\Theta = \Omega\left(1 - \frac{\xi^2}{6}\,\frac{1-\omega^2\tau_B^2}{\left(1+\omega^2\tau_B^2\right)^2}\right), \tag{SM-33}$$

which is (SM-33) in corrected form. ■

47.3 Physical Interpretation: the Whole Effect in One Line

The authors' commentary on this formula is the clearest statement of the physics in either paper, and it deserves to be reproduced in our own words and then checked against the algebra.

Permanent or slowly varying field, $\omega\tau_B < 1$. Then $1-\omega^2\tau_B^2 > 0$, the bracket is less than unity, and

$$\Theta < \Omega. \tag{47.3.1}$$

The particles rotate *slower* than the surrounding fluid. The liquid is forced to flow past them, extra kinetic energy is dissipated, and the additional viscosity is positive.

Fast alternating field, $\omega\tau_B > 1$. Then $1-\omega^2\tau_B^2 < 0$, the bracket exceeds unity, and

$$\Theta > \Omega. \tag{47.3.2}$$

The particles rotate *faster* than the fluid. They no longer retard the flow; they accelerate it. And $\Delta\eta < 0$. Shliomis & Morozov emphasise the reversal: instead of retarding the flow the grains now accelerate it, so that $\Delta\eta$ turns negative — the natural consequence, they observe, of part of the alternating field's energy being converted into kinetic energy of the liquid.

The two extreme cases as checks.

- *No field.* $\xi = 0$ gives $\Theta = \Omega$ exactly: the particles turn with the fluid and $\Delta\eta = 0$, as it must be.
- *Strong stationary field.* The formula above is a weak-field result, but its trend is right: as ξ grows at $\omega = 0$, Θ falls. The exact statement is the one derived in Volume II: in a saturating stationary field the moments are locked, $\Theta \to 0$, and $\Delta\eta \to \frac{3}{2}\eta\varphi$ — Eq. (SM-4).

47.4 Connection to the Next Equation

Comparing (SM-33) with the master formula closes the logical circle. That is (SM-34).

Chapter 48

Equation (SM-34): Closing the Circle

$$\Delta\eta = \frac{3}{2}\,\eta\varphi\,\frac{\Omega-\Theta}{\Omega}. \tag{SM-34}$$

48.1 Context

The authors write: "With the help of (33), expression (28) may be transformed into (34)." In this book the relation was derived long ago, in Sec. 14.4, as the general bridge between particle spin and viscosity; here it reappears as a *consequence* of the explicit solution, which is a nontrivial check that no factor has been lost anywhere in Volumes IV and V.

48.2 Step-by-Step Derivation

Step 1: form the slip from (SM-33). Subtracting (47.2.3) from $\boldsymbol{\Omega}$:

$$\boldsymbol{\Omega}-\boldsymbol{\Theta} = \boldsymbol{\Omega}-\boldsymbol{\Omega}\left(1-\frac{\xi^2}{6}\cos^2\alpha\cos 2\alpha\right) = \frac{\xi^2}{6}\cos^2\alpha\cos 2\alpha\,\boldsymbol{\Omega}. \tag{48.2.1}$$

Step 2: divide by Ω and multiply by $\frac{3}{2}\eta\varphi$.

$$\frac{3}{2}\eta\varphi\,\frac{\Omega-\Theta}{\Omega} = \frac{3}{2}\eta\varphi\cdot\frac{\xi^2}{6}\cos^2\alpha\cos 2\alpha \tag{48.2.2}$$

$$= \frac{3\eta\varphi\xi^2}{12}\cos^2\alpha\cos 2\alpha \tag{48.2.3}$$

$$= \frac{1}{4}\eta\varphi\xi^2\cos^2\alpha\cos 2\alpha. \tag{48.2.4}$$

Step 3: recognise the right-hand side. This is precisely (SM-27), which by Chapter 42 equals (SM-28). Hence

$$\Delta\eta = \frac{3}{2}\eta\varphi\frac{\Omega-\Theta}{\Omega}, \tag{SM-34}$$

■

48.3 Why This Is More Than an Identity

Equation (SM-34) was derived in Volume II from angular-momentum bookkeeping alone, with *no* reference to the magnetisation dynamics, the field waveform, or the weak-field approximation. Its rederivation here from the explicit weak-field solution therefore checks the entire chain

$$\text{(SM-16)} \to \text{(SM-23)} \to \text{(SM-25)} \to \text{(SM-27)} \to \text{(SM-28)} \to \text{(SM-33)} \to \text{(SM-34)}.$$

A single misplaced factor of 2, or a $\cos^2\alpha$ where a $\cos 2\alpha$ belonged, would break the identity.

Its deeper significance is stated by the authors themselves in Sec. III: because (SM-34) rests only on the torque balance, a formula obtained under the weak-field assumption in fact holds for any field strength and any frequency. That is the bridge into the finite-amplitude theory, which will occupy Volume VI: the task there is not to redo the viscosity calculation but merely to compute $(\Omega - \Theta)/\Omega$ — equivalently the averaged torque — without the weak-field assumption.

Chapter 49

Status and Road Map

49.1 What has been established in Volume V

Equation	Statement	Where derived
(SM-25)	$\boldsymbol{M} \times \boldsymbol{H} = -2\Omega\eta\varphi\xi^2 \cos^2\alpha(\cos^2\omega t \cos 2\alpha + \sin\omega t \cos\omega t \sin 2\alpha)$	Ch. 39
(SM-26)	averaged momentum equation; two-scale expansion	Ch. 40
(41.2.3)	$\overline{\boldsymbol{M} \times \boldsymbol{H}} = -\eta\varphi\xi^2 \cos^2\alpha \cos 2\alpha\, \boldsymbol{\Omega}$	Sec. 41.2
(SM-27)	$\Delta\eta = \frac{1}{4}\eta\varphi\xi^2 \cos^2\alpha \cos 2\alpha$	Ch. 41
(SM-28)	$\Delta\eta = \frac{1}{4}\eta\varphi\xi^2 \dfrac{1-\omega^2\tau_B^2}{(1+\omega^2\tau_B^2)^2}$; zero at $\omega\tau_B = 1$, minimum $-\Delta\eta(0)/8$ at $\omega\tau_B = \sqrt{3}$	Ch. 42
(SM-29)	$\Delta\eta = -(\varphi/\eta)(M_s H/6\omega)^2$; τ_B cancels	Ch. 43
(SM-30)	$M_x = \chi H_0 \cos\alpha \cos(\omega t - \alpha)$, second variant	Ch. 44
(SM-31)	coupled equations for a and θ	Ch. 45
(SM-32)	$\theta = 2\alpha$, $a = \Omega\tau_B\chi H_0 \cos^2\alpha$; agrees with (SM-23)	Ch. 46
(SM-33)	$\Theta = \Omega[1 - \frac{\xi^2}{6}(1-\omega^2\tau_B^2)/(1+\omega^2\tau_B^2)^2]$	Ch. 47
(SM-34)	$\Delta\eta = \frac{3}{2}\eta\varphi(\Omega - \Theta)/\Omega$, recovered	Ch. 48

49.2 Cumulative progress

Shliomis & Morozov: Eqs. (SM-1)–(SM-34) complete — the whole of Secs. I and II of the paper. Bacri *et al.*: Eqs. (B-1)–(B-4) and (B-10) complete.

49.3 What remains

- **Volume VI** — Eqs. (SM-35)–(SM-43), Sec. III of the paper: the nonlinear magnetisation equation at arbitrary amplitude; the decomposition $\boldsymbol{M} = (nmF,\ nm\Omega\tau_B G,\ 0)$; the coupled equations (SM-36) for F and G; the rectangular waveform $\gamma(t)$; the averaged torque (SM-37); the general formula (SM-38)–(SM-39) and its reduction to (SM-34); the analytic weak-field result (SM-40) for square-wave modulation; the half-period identity (SM-41); the formal solution (SM-42)–(SM-43); and the numerical results of Figs. 3–5,

including the values $\omega_0\tau_B = 0.98$, $\omega_m\tau_B = 1.707$, $\Delta\eta_m = -0.105\,\Delta\eta(0)$ for the rectangular case.

- **Volume VII** — Eqs. (B-5)–(B-9): the effective-field method of Martsenyuk, Raikher and Shliomis derived from the Smoluchowski equation (11.6.1); the equation (B-6) for $\boldsymbol{\xi}_e$; the zero-approximation equation (B-7); the function F of (B-8); the averaged torque and the function g of (B-9); and the comparison with the measured isolines of Figs. 2 and 5, including the critical values $\xi_c = 126$, $\omega_c\tau_B = 20$ for $\eta_{\text{total}} = 0$.

End of Volume V. Volume VI begins with Eq. (SM-35).